Crucis: An Independent Framework for Auditing Causal-Discovery Systems

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Abstract

A causal discovery system may use numerical patterns, variable names, or both to decide which arrows to draw. A final benchmark score tells us little about how it reached those decisions. We introduce Crucis, an independent framework for auditing externally developed systems by changing one source of information at a time where the comparison permits, and following changes through recorded stages. The contracts for connecting systems and defining experiments are implemented and tested. Recorded audits illustrate what the approach can reveal. On the CausalArena semantic cases we examined, using column order to settle uncertain arrow directions raised the Fast Causal Inference algorithm's score on identical graphs. Changing variable names while keeping the data fixed lowered direction scores in several recorded comparisons. These used different measurement setups, and the Causal-Copilot result was exploratory. In a separate analysis of completed Causal-Copilot runs, the largest paired drop in the score for which variables were connected occurred during bootstrap refinement, before language-model review.

Framework: https://github.com/crucis-lab/crucis
Research evidence: https://github.com/uzairgheewala/cc-research

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1 Introduction

If we change one variable, what else moves? When an experiment is impractical, causal discovery can propose a graph from the available data. Language models can now participate in building that graph. Causal-Copilot uses one to inspect data, help select and configure an algorithm, review the graph, and report the result (Wang et al. 2025). Other systems use a model to propose constraints, a causal order, or edits to a data-driven graph (Khatibi et al. 2024; Shen et al. 2025; Vashishtha et al. 2025).

1.1 What should an audit tell us?

The same score can reflect patterns in the data or reliance on clues in the input. Variable scales can reveal causal order in simulated benchmarks (Reisach, Seiler and Weichwald 2021). Column order can affect a search algorithm's output (Kitson and Constantinou 2024), and language models can reconstruct familiar graphs from names alone (Srivastava et al. 2025). An independent audit should make these sources distinguishable. Crucis is designed to do that across externally developed systems. Causal-Copilot is the worked example for tracing changes inside one system.

1.2 How does Crucis examine a result?

A connection records what a system receives and returns. An experiment then states what changes, what stays fixed, and how outputs will be compared. If intermediate records are available, the audit can locate where a change appeared. Each level answers a different question.

L0 **Score the graph.** Compare the output with the stated reference.
L1 **Change the input clues.** Keep the numbers fixed while changing names, column order, or file paths under the specified controls.
L2 **Trace the changes.** Use available stage records to locate added, removed, or reversed links. Link them to logged operations where the record permits.
L3 **Trace claims.** The design calls for tracing individual claims back to their sources.
L4 **Search for failures.** The design calls for adversarial tests of where a system's claims break down.

The system connection and experiment definition contracts are implemented and tested. Execution and collection remain in design, as do the L3 and L4 capabilities described here. The recorded audits exercise parts of the design on retained evidence. Native execution across systems, validation of the auditor, and the broader multi-system study remain ahead.

An audit reports what information the system had, such as data alone or data with interventions. It also reports what the comparison can establish. A graph used to simulate data can support an accuracy comparison. When published reference graphs disagree, the score

measures agreement with the specified version. Section 3 explains which reference each audit uses.

1.3 What have the recorded audits shown?

The audits cover 5 language-model systems and the CausalArena and CausalBench benchmarks. Each example below answers a different question.

Can the scoring rule change the result? On the evaluated CausalArena semantic subset, using column order to settle uncertain directions raised the directed F1 score of Fast Causal Inference (FCI) by 0.221 relative to averaging over those directions on the same graphs. See Appendix A.1 for the scoring comparison.

Can the names change the result? With numerical data fixed, swapping names lowered directed F1 by at least 0.10 in 3 of 4 eligible entries. The entries use different cohorts, scoring rules, and output stages, including an earlier exploratory Causal-Copilot comparison. Appendix A.2 explains how to read the tally.

Where did the graph change? In the completed Causal-Copilot main-control runs, the largest paired loss in adjacency F1 occurred during bootstrap refinement, before language-model review. Appendix A.3 follows the relevant stages and accounts for crashes.

These results show why the pieces of an audit belong together. The scoring example depends on holding the graphs fixed. The name comparison depends on holding the data fixed and knowing which output was measured. The bootstrap result depends on following the same completed runs through their intermediate graphs. A final score alone cannot make those distinctions.

2 Prior work

2.1 How causal discovery is evaluated, and what benchmarks reward

Greedy Equivalence Search compares classes of graphs using a score and can identify the generating structure under the conditions of its large-sample result (Chickering 2002). Benchmarks with known graphs provide a way to check recovery, but the inputs can reveal features of the reference graph (Reisach, Seiler and Weichwald 2021; Kitson and Constantinou 2024). In common simulated settings the variance of the variables grows along the causal order, so sorting by variance matches continuous-optimisation methods (Reisach, Seiler and Weichwald 2021), and a scale-invariant version of the same artefact is also high in common settings and can be exploited (Reisach et al. 2023). The order of the columns in the input data alone changes the graph that hill-climbing search learns, typically more than a change in sample size does (Kitson and Constantinou 2024). Real-data evaluation leans on very few datasets, above all the Sachs protein-signalling data, whose reference graph is not definitive (Brouillard et al. 2025; Sachs et al. 2005). CausalArena collects syn-

thetic, semantic and formula-grounded models with a public leaderboard (Li et al. 2026), and CausalBench scores gene-network inference on held-out perturbation data (Chevalley et al. 2025; Replogle et al. 2022).

2.2 Language models as a source of causal knowledge

Language models answer pairwise causal questions about named variables well, and part of that skill is memorised (Kiciman et al. 2023); they may repeat causal statements without performing causal inference (Zečević et al. 2023). When a model is treated as an imperfect expert, the guarantee that the true graph is kept holds only under an assumed error model, and in the authors' own case study the model was poorly calibrated and the true graph was excluded on one benchmark (Long et al. 2023). Asking for a causal order instead of individual edges avoids confusing direct with indirect effects (Vashishtha et al. 2025). Controls for memorisation now exist: neutral aliases in the model's queries (Kitson and Constantinou 2026), graphs published after a model's training cut-off (Srivastava et al. 2025), random graphs carrying real concept names (Li and Russo 2026), and names moved onto the wrong columns with the data unchanged, under which models' accuracy collapses (Dong et al. 2025). Our audit battery uses wrong-name reassignment alongside separate layout controls where the recorded evidence supports those comparisons.

2.3 Causal agents and how they are evaluated

Causal-Copilot automates the whole analysis and states that the model's recommendations are soft suggestions that may not override the high-confidence decisions of its bootstrap procedure (Wang et al. 2025). ALCM, MATMCD and Causal Modelling Agents let a model refine or constrain a data-driven graph (Khatibi et al. 2024; Shen et al. 2025; Abdulaal et al. 2024), and open libraries expose model-based causal suggestions (CausalIQ 2026; PyWhy 2026). CausalSteward evaluates Causal-Copilot as a baseline (Tagliapietra et al. 2026). Repeated runs of the same agent can disagree, so single-run comparisons can misrank systems (Mehta 2026). A position paper argues that agents should not supply edges or causal conclusions at all and asks for audit trails (Zheng et al. 2026).

Related benchmarks and auditing frameworks. Heuresis's PC-centred trace decomposition offers a precedent for L2, alongside instrument checks and a name-only leakage probe in its interactive evaluation on anonymous, randomly ordered simulations (Ma, Ding and Wang 2026). ACDB separates observational recovery, DAG recovery and intervention efficiency in a benchmark with budgeted interventions, and tests how graph density affects a structure-blind baseline (Deb, Shah and Krishnan 2026). TCD-Arena provides a modular test kit for robustness to controlled time-series assumption violations (Stein et al. 2026). Causal-Audit uses time-series assumption diagnostics and calibrated risk to recommend a method or abstain before discovery within its evaluated decision pipeline (Ruiz et al. 2026). The design question for Crucis is how to use the records an external system can supply to

make a stage comparison, and how to report where those records leave the comparison unresolved.

2.4 How much an edge can be trusted

Bootstrap resampling gives confidence measures for features of a learned network (Friedman, Goldszmidt and Wyner 1999), but the resulting edge frequencies can be miscalibrated (Kummerfeld and Rix 2019; Banerjee, Andrews and Kummerfeld 2025). Language models asked about direct edges give high stated confidence to most of their false-positive edges; agreement across prompts is better calibrated, though not significantly so (Kumar et al. 2026). An audit therefore has to test whether a system's confidence means anything, not assume it.

2.5 Auditing practice the protocol borrows

Metamorphic testing checks a program that has no oracle for each output by checking relations between the outputs of related inputs (Chen, Cheung and Yiu 1998; Segura et al. 2016); renaming or reordering columns while holding the numbers fixed is such a relation. Behavioural testing of language-processing models crosses capabilities with test types that include invariance tests, and finds failures that held-out accuracy hides (Ribeiro et al. 2020). Model cards and datasheets document a model's evaluation and a dataset's origin (Mitchell et al. 2019; Gebru et al. 2021), internal algorithmic audits define staged audit artefacts (Raji et al. 2020), and PROV-DM records which agent and activity produced a piece of data (Moreau and Missier 2013). Our certificate combines these ideas for causal claims: system identity, battery hash, every number with its source, and the command that reruns it.

3 Data

The audit draws on a suite built from CausalArena's semantic scenarios, Sachs and 4 other named datasets that ship with Causal-Copilot, and CausalBench's perturbation data. Table 1 shows what each audit uses as a reference, from a known simulated graph to held-out interventions. Shapes come from suite manifests and retained CausalBench filtering metadata (Appendix A.4). The checkpoint binds these sources by hash.

CausalArena semantic suite. CausalArena releases structural causal models whose variables carry names from a written scenario, with observational exports (Li et al. 2026). We take the 10 semantic scenarios of our test split, the first replicate of the observational-only release, 1,000 rows each. Each scenario is written 3 times with byte-identical numbers: under its real names, under neutral names, and under its own names moved onto the wrong

Table 1: Selected inputs by audit. The semantic rows describe the Causal-Copilot name battery. For CausalBench, cells and perturbed genes are counted after filtering; held-out interventions provide an evaluation regime without a reference graph. Other tiers describe graph reliability (Section 1).

Group	Dataset	Rows	Variables	Ref. edges	Tier	Licence
CausalArena	10 semantic scenarios	1,000	20–25	35–58	exact	CC BY 4.0
Named	Sachs	853	11	19	contested	none stated
	Abalone	4,177	8	7	provenance	CC BY 4.0
	Auto MPG	392	5	5	mismatched	CC BY 4.0
	Concrete strength	1,030	9	8	provenance	CC BY 4.0
	DWD climate	349	5	4	provenance	see text
CausalBench	K562	162,751	622	–	held-out interventions	CC BY 4.0
	RPE1	162,733	383	–	held-out interventions	CC BY 4.0

columns by a derangement chosen to reduce pairs adjacent under both readings (30 tables). Because the released column order is a causal order, the columns are shuffled with a fixed seed, and the file paths are opaque codes. The reference is the graph the data were simulated from, so scores are accuracy. The layout audit of the public leaderboard uses CausalArena's public files instead (250 semantic, 250 formula and 100 synthetic). The dataset is released under CC BY 4.0 on Hugging Face (`LAMDA-Tabular/CausalArena`).

Named datasets. Causal-Copilot ships Sachs and 4 other tables with reference graphs, and we use its copies (Wang et al. 2025). Sachs (Sachs et al. 2005) has 4 arms (the original, a copy under another path, neutral names, and moved names). Its reference is contested: the repository's graph has 19 edges, the bnlearn version 17, sharing 16 adjacencies with 1 reversed, and CausalArena uses a version with 20 edges, so a score on Sachs is agreement with one version, not accuracy. Abalone (Nash et al. 1994), Auto MPG (Quinlan 1993) and Concrete compressive strength (Yeh 1998) come from the UCI repository under CC BY 4.0, and the climate table matches the Deutscher Wetterdienst pairs of the Tübingen cause-effect benchmark (Mooij et al. 2016). The DWD open-data terms are CC BY 4.0; the copy's lineage is inferred from its match to the benchmark pairs. Each has 3 arms (real, neutral, swapped). Their reference graphs are expert graphs with nothing to compare against (provenance tier), except Auto MPG, whose reference differs from CausalArena's version (4 variables, 3 edges, 398 rows), so its scores must not be set beside CausalArena's row. The checked sources did not identify a licence for the Sachs data.

CausalBench. CausalBench uses the CRISPRi Perturb-seq screens of Replogle et al. (2022) in the K562 and RPE1 cell lines and scores a predicted network on held-out interventions by a mean Wasserstein distance and a false omission rate (Chevalley et al. 2025). In the retained metadata for CausalBench's filtering, the K562 data have 162,751 cells (10,691 unperturbed controls) and 622 perturbed genes, and RPE1 162,733 cells (11,485 controls) and 383 genes. The screens are released under CC BY 4.0 and the benchmark code un-

der Apache 2.0. Our audit of this benchmark is cited from its result file. All 4 published MeanDifference rows match within the recorded tolerances. At the smaller evaluated edge budget, MeanDifference reaches 0.90–0.95 of a test-informed reference that ranks pairs by knockdown effects measured on the test split. Its false omission rate stays within the audit's declared margin of the matched random-network mean in 3 of the 4 cells. The random-network comparison was unavailable in the other cell.

Recorded runs. The instrumented Causal-Copilot series records code identity, the model snapshot, model requests and replies, and stage graphs and refinement steps as captured by the run. Causal-Copilot was run unmodified (upstream commit `a31aa7c`) with gpt-4o-2024-08-06: 30 runs on the semantic suite in the name-arm study, and a main-control series of 30 runs on the same tables (API cost $12.57). The other systems' name-control results use a different semantic cohort and their own declared output/scoring contracts. Their results are not paired scenario-for-scenario with the Causal-Copilot collection.

Limits of the data. The semantic scenarios are simulated, so their exactness comes with simulation's artefacts; each table and arm was run once; one model family was used; and the named datasets' references are expert graphs. The protocol treats each of these as a declared regime or tier rather than hiding it.

Code

The reusable framework is published at `https://github.com/crucis-lab/crucis`, with documentation at `https://crucis-lab.github.io/crucis/`. The companion repository (`https://github.com/uzairgheewala/causal-copilot-q1`) uses retained graphs to recompute its Causal-Copilot run comparison with one command and no API key. The stage-specific attribution uses separate evidence in the research repository (`https://github.com/uzairgheewala/cc-research`). Appendix A.4 maps reported values to result files, retained diagnostic snapshots and separately preserved CausalBench metadata. The numerical evidence is pinned to `aef649241abe`.

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Appendices

A.1 Scoring the same graphs A1
A.2 Changing names while holding the data fixed A1
A.3 Locating a change inside a pipeline A2
A.4 Where every number comes from A3

Use this appendix to follow an audit from its question to the evidence behind the answer. Start with scoring (Appendix A.1), names (Appendix A.2), or pipeline stages (Appendix A.3). Appendix A.4 is the lookup for exact values and their sources.

A.1 Scoring the same graphs

CausalArena's released files list the variables in causal order. The scoring rule uses column position to settle directions the algorithm left uncertain. On the evaluated semantic subset, FCI gains 0.221 directed F1 under that rule relative to expectation scoring on the same graphs. The published PC row is 0.333; a distinct PC maxP variant scores 0.463 on shuffled columns under expectation scoring. The FCI and maxP rows cover subsets of the public inventory described in Section 3.

The separate CausalBench comparison in Section 3 asks what the metrics reward when a reference graph is unavailable. Its smaller edge budget, test-informed reference and unavailable random-network comparison limit what can be inferred.

A.2 Changing names while holding the data fixed

The name control pairs graphs produced from the same numerical table under real and swapped variable names. The AB-06 tally applies a 5-pair entry floor to each system's declared output stage. PyWhy-LLM falls below that floor.

Swapping names lowers directed F1 by at least 0.10 in 3 of 4 entries. The drops are 0.251 for causaliq-knowledge, 0.167 for Causal-Copilot, and 0.122 for MATMCD. The pipeline of Vashishtha et al. (2025) shows a drop of 0.020 under its tested conditions.

The tally describes sensitivity under each entry's measurement. The earlier exploratory Causal-Copilot comparison scores committed marks in the graph handed to cycle removal. The other agents use final outputs and expectation scoring on a semantic cohort with no shared scenarios. The tally therefore cannot rank these systems under common conditions.

Which reading do the retained changes follow? Inside Causal-Copilot, BND-08's recorded label-following calculation counts distinct (scenario, directed-edge) pairs retained in completed swapped-name outputs, changed from bootstrap and matching their post-review pair state at final output, and adjacent in exactly one reference reading. Among these 101 informative pairs, 81 are adjacent under the label reading and 20 under the data reading. This classification tests adjacency, not agreement of edge direction; edges adjacent in both references or neither are outside this denominator. The recorded decision rule comes from BND-08's pre-run specification, not a later finding threshold. However, that specification describes a pre-cycle endpoint while the recorded calculation consumes completed final outputs. We retain the counts as a descriptive diagnostic, without claiming confirmation of the specified endpoint. Failed swapped runs cannot enter this completed-output calculation. Edges from the same scenario are dependent. The preserved manifest binds the rule's bytes. It does not establish when the rule became public or reconcile the different endpoints.

A.3 Locating a change inside a pipeline

Causal-Copilot's main-control initial mean adjacency F1 is 0.584 across all 30 tables. Its final mean is 0.321 among the 22 completions. In paired diagnostics on those completions, the largest observed stage loss occurs at bootstrap within refinement, before model review. The initial and final means above describe different populations. The stage comparison instead follows the same completed runs through successive graphs. A real-arm-only effect answers a different question and is not substituted here.

The completed-run comparison leaves out a separate failure mode. All 8 runs that reached cycle removal crashed there. The logs record a key error in 8 of those entrants.



Figure A 1: The bootstrap transition had the largest paired drop in adjacency F1 among the completed Causal-Copilot main-control runs. The shaded stage locates that transition before language-model review.

A separate comparison against a data-only graph. In the real-name MATMCD runs, pooled precision among committed claims on the released intervention targets is lower for the final output than for the data-only variant on the point estimate (0.6855 against 0.7199); the interval for the difference includes zero. The pre-registered point rule applies to the estimate.

A.4 Where every number comes from

Generated by `docs/paper/q1/build_numbers.py` from `numbers.json`. Paths are relative to the root of cc-research. Selectors extend JSON pointers: `*` takes every child and `[k=v]` selects the first matching list item. Source flags identify inputs outside the tracked build base or with differences from it. The CausalBench filtering metadata are separately preserved and hashed. The table locates evidence for each value; the companion replay covers only its stated comparison.

Key	Value	Source file and JSON pointer (cc-research)
`cc.arms`	30	`analysis/results/fork/bld06_live_compare.json /arms`
`cc.completed`	22	`analysis/results/fork/bld06_live_compare.json /completed/main`
`cc.s0-adj`	0.584	`analysis/results/fork/bld06_live_compare.json /means/main/S0_adj_f1`
`cc.final-adj`	0.321	`analysis/results/fork/bld06_live_compare.json /means/main/final_adj_f1`
`cc.usd`	$12.57	`analysis/results/fork/bld06_live_compare.json /usd/main`
`cc.cyc.entered`	8	`analysis/results/fork/bld06_cycles.json /arms/main/runs_entering_cycle_step`
`cc.cyc.keyerr`	8	`analysis/results/fork/bld06_cycles.json /arms/main/cycle_step_keyerrors`
`bnd08.runs`	30	len of: `analysis/results/stability_s2/bnd08_summary.json /runs`
`bnd08.m3.labels`	81	`analysis/results/stability_s2/bnd08_summary.json /M3/labels_only`
`bnd08.m3.data`	20	`analysis/results/stability_s2/bnd08_summary.json /M3/data_only`
`bnd08.m3.n`	101	`analysis/results/stability_s2/bnd08_summary.json /M3/n`
`bnd08.model`	gpt-4o-2024-08-06	`analysis/results/stability_s2/bnd08_summary.json /snapshots/0`
`ab06.agents`	4	len of: `analysis/results/ab06/ab06.json /agents`
`ab06.cc-entered`	1	count_true of: `analysis/results/ab06/ab06.json /causal_copilot/*/counted_in_P9`
`ab06.systems`	5	add of: key `ab06.agents`; key `ab06.cc-entered`
`ab06.p9.entered`	4	len of: `analysis/results/ab06/ab06.json /P9/systems_entered`
`ab06.p9.losing`	3	len of: `analysis/results/ab06/ab06.json /P9/losing_ge_0.10`
`ab06.p9.min-scen`	5	`analysis/results/ab06/ab06.json /P9/min_paired_scenarios`
`ab06.p9.threshold`	0.10	`plans/ab06_prereg.json /predictions/P9/supported` (pattern `lose ≥ ([0-9.]+)`)

Key	Value	Source file and JSON pointer (cc-research)
ab06.causaliq.drop	0.251	analysis/results/ab06/ab06.json /agents/causaliq/name_reliance_real_minus_swapped/mean
ab06.cc.drop	0.167	analysis/results/ab06/ab06.json /causal_copilot/causal_copilot_gpt4o/name_reliance_real_minus_swapped/mean
ab06.matmcd.drop	0.122	analysis/results/ab06/ab06.json /agents/matmcd/name_reliance_real_minus_swapped/mean
ab06.vashishtha.drop	0.020	analysis/results/ab06/ab06.json /agents/vashishtha/name_reliance_real_minus_swapped/mean
ab06.matmcd.dv1	0.6855	analysis/results/ab06/ab06.json /agents/matmcd/data_only_floor_real_arm/dv1_precision_final
ab06.matmcd.dv1-data	0.7199	analysis/results/ab06/ab06.json /agents/matmcd/data_only_floor_real_arm/dv1_precision_data_only
ab01.pc.published	0.333	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=PC (upstream, default rule);family=semantic]/published
ab01.pcmaxp.controlled	0.463	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=PC (upstream, maxP);family=semantic]/ours_shuffled_expectation
ab01.fci.authors	0.578	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=FCI;family=semantic]/ours_released_authors_rule
ab01.fci.expect	0.357	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=FCI;family=semantic]/ours_released_expectation
ab01.fci.gain	0.221	sub of: key ab01.fci.authors; key ab01.fci.expect
ab01.files.semantic	250	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=DirectLiNGAM;family=semantic]/files
ab01.files.formula	250	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=DirectLiNGAM;family=formula]/files
ab01.files.synthetic	100	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=DirectLiNGAM;family=synthetic]/files
ben13.reproduced	4	analysis/results/ben13/summary.json /reproduced_rows
ben13.cells	4	count_notnull of: analysis/results/ben13/summary.json /lines/*/controls/*/placement_share_wasserstein

Key	Value	Source file and JSON pointer (cc-research)
ben13.for.random-level	3	count_notnull of: analysis/results/ben13/summary.json /lines/*/controls/*/?placement_share_for_note
ben13.echo.min	0.90	min of: analysis/results/ben13/summary.json /predictions/P3/values/*
ben13.echo.max	0.95	max of: analysis/results/ben13/summary.json /predictions/P3/values/*
s2.families	10	len of: datasets/suite_q1s2/MAPPING.json /families
s2.tables	30	len of: datasets/suite_q1s2/MAPPING.json /arms
s2.arms-per	3	div of: key s2.tables; key s2.families
s2.rows	1,000	max of: datasets/suite_q1s2/MAPPING.json /families/*/rows
s2.vars.min	20	min of: datasets/suite_q1s2/MAPPING.json /families/*/variables
s2.vars.max	25	max of: datasets/suite_q1s2/MAPPING.json /families/*/variables
s2.edges.min	35	min of: datasets/suite_q1s2/MAPPING.json /families/*/edges
s2.edges.max	58	max of: datasets/suite_q1s2/MAPPING.json /families/*/edges
s1.sachs.rows	853	datasets/MAPPING.json /source/rows
s1.sachs.vars	11	len of: datasets/MAPPING.json /source/columns
s1.sachs.edges	19	len of: datasets/MAPPING.json /source/reference_edges
s1.sachs.arms	4	len of: datasets/MAPPING.json /arms
s1.sachs.bnlearn	17	analysis/results/reference_reliability/reliability.json /families/sachs/repo_vs_bnlearn/edges_b
s1.sachs.shared	16	analysis/results/reference_reliability/reliability.json /families/sachs/repo_vs_bnlearn/shared_adjacencies
s1.sachs.reversed	1	analysis/results/reference_reliability/reliability.json /families/sachs/repo_vs_bnlearn/reversed
s1.sachs.arena	20	analysis/results/reference_reliability/reliability.json /families/sachs/causalarena_version/edges
s1.abalone.rows	4,177	datasets/suite_b1/MAPPING.json /families/abalone/rows
s1.abalone.vars	8	len of: datasets/suite_b1/MAPPING.json /families/abalone/columns
s1.abalone.edges	7	len of: datasets/suite_b1/MAPPING.json /families/abalone/reference_edges
s1.autompg.rows	392	datasets/suite_b1/MAPPING.json /families/autompg/rows

Key	Value	Source file and JSON pointer (cc-research)
s1.autompg.vars	5	len of: datasets/suite_b1/MAPPING.json /families/autompg/columns
s1.autompg.edges	5	len of: datasets/suite_b1/MAPPING.json /families/autompg/reference_edges
s1.concrete.rows	1,030	datasets/suite_b1/MAPPING.json /families/concrete/rows
s1.concrete.vars	9	len of: datasets/suite_b1/MAPPING.json /families/concrete/columns
s1.concrete.edges	8	len of: datasets/suite_b1/MAPPING.json /families/concrete/reference_edges
s1.dwd.rows	349	datasets/suite_b1/MAPPING.json /families/dwd/rows
s1.dwd.vars	5	len of: datasets/suite_b1/MAPPING.json /families/dwd/columns
s1.dwd.edges	4	len of: datasets/suite_b1/MAPPING.json /families/dwd/reference_edges
s1.autompg.arena.vars	4	analysis/results/reference_reliability/reliability.json /families/autompg/causalarena_vars_edges_rows/0
s1.autompg.arena.edges	3	analysis/results/reference_reliability/reliability.json /families/autompg/causalarena_vars_edges_rows/1
s1.autompg.arena.rows	398	analysis/results/reference_reliability/reliability.json /families/autompg/causalarena_vars_edges_rows/2
s1.named.tables	12	len of: datasets/suite_b1/MAPPING.json /arms
s1.named	4	len of: datasets/suite_b1/MAPPING.json /families
s1.named.arms-per	3	div of: key s1.named.tables; key s1.named
cb.k562.cells	162,751	external/causalbench/dataset_k562_filtered.npz.json /cells_out *(excluded)*
cb.k562.genes	622	external/causalbench/dataset_k562_filtered.npz.json /genes_out *(excluded)*
cb.k562.controls	10,691	external/causalbench/dataset_k562_filtered.npz.json /controls *(excluded)*
cb.rpe1.cells	162,733	external/causalbench/dataset_rpe1_filtered.npz.json /cells_out *(excluded)*
cb.rpe1.genes	383	external/causalbench/dataset_rpe1_filtered.npz.json /genes_out *(excluded)*
cb.rpe1.controls	11,485	external/causalbench/dataset_rpe1_filtered.npz.json /controls *(excluded)*